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Review

Safeguarding rice crops in Nepal: unveiling strategies against the yellow stem borer (*Scirpophaga incertulas*)

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Abstract
Rice (*Oryza sativa* L.) is a vital crop with enormous economic and nutritional importance in Nepal and worldwide. The yellow stem borer (YSB), *Scirpophaga incertulas* (Walker) (Lepidoptera: Crambidae), is a monophagous severe pest that poses a substantial threat to rice crops, causing significant yield losses by its larva feeding on stems and tillers. This review covers the life cycle, distribution, damage, and management and control strategies of YSB. YSB primarily targets rice plants, causing symptoms like dead hearts and whiteheads. Integrated pest management (IPM) strategies are crucial for effective YSB control. Cultural practices include selecting resistant rice varieties, using *Bacillus thuringiensis* (Bt) incorporated transgenic rice plants, synchronizing planting to reduce generations, and promoting natural enemies by conserving their habitats. Mechanical methods involve pheromone and light traps to disrupt mating and reduce adult populations. Biological control methods utilize natural predators like beetles and spiders and parasitoids such as *Cotesia flavipes* and *Trichogramma* spp. In contrast, chemical control involves targeted insecticide application when pest populations reach economic threshold levels that is when 5–10% dead hearts are seen. The article emphasizes the importance of IPM techniques in reducing YSB infestation and fostering sustainable rice production. These approaches not only help to lower the YSB population but also contribute to environmental preservation and long-term agricultural viability. As rice is a staple food for millions, effective management of YSB becomes crucial for guaranteeing food security and livelihoods across the globe.

Keywords Yellow stem borer · Pest · Life cycle · Rice · Damage · Management

1 Introduction

Agriculture alone contributes 24.26% GDP of Nepal and is considered the backbone of the Nepalese economy. In many countries of South Asia including Nepal, rice is regarded as the largest crop industry and plays a significant role in agricultural and economic development [1]. Rice ranks in the second position after wheat among the most cultivated cereals in the world [2]. Rice is the top major cereal crop grown in Nepal in terms of area, and production because of its high importance as a staple food and its straw need for animal feed. The role of rice in global food security is unavoidable in the present condition and future as well [3]. It is the most preferred staple food crop of Nepal and fulfills more than 50% of the calorie requirement of Nepalese people. According to Krishi Diary 2023, the total yield of rice in 2020/21 was 5,550,878 Metric tons in 1,458,915 hectares of area. In 2021/22, the yield of rice was

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5,621,710 Metric tons in 1,473,474 hectares of land, and in 2022/23, the yield was 5,130,625 metric tons in 1,477,378 hectares of land [4]. This means, the productivity of rice in 2020/21 was 3.804 Mt/ha, 3.81 Mt/ha in 2021/22, and 3.473 Mt/ha in the year 2022/23. Although high-yielding cultivars resistant to climate change and advancements in technology have contributed to an increase in rice production, the crop's actual yield remains 10–15% below its potential owing to biotic and abiotic stress. The most severe abiotic stressors that affect rice crops include drought, excessive temperatures, salt, hazardous radiation, heavy metals, and gaseous pollution [5]. The biotic stress factors include various insects i.e., stem borer, gall midge, brown plant hopper, and diseases caused by fungi, viruses, bacteria, and nematodes [6].

The yellow stem borer (YSB) *Scirpophaga incertulas* (Lepidoptera: Crambidae) [7], is regarded as the major pest in deep water rice in Asian countries where it causes yield loss estimated at the rate of 15–20%. They can bore into the sheath and destroy crops. YSB at its larval stage bores into the stem and causes dead hearts during the vegetative stage which leads to loss of productive tillers and white ear damage at the maturity phase of rice resulting in a huge economic loss. They can cause yield losses ranging from 10 to 60% [8].

This review is essential due to the critical importance of rice in Nepal's agriculture. The relentless damage caused by YSB to rice crops demands a comprehensive review regarding the biology of yellow stem borer and its management strategies. This review not only highlights the need for sustainable and eco-friendly approaches to control YSB, including IPM but also underscores the importance of selecting resistant varieties, cultural practices, and conservation of natural enemies. This review aids in safeguarding rice crops in Nepal by protecting yield loss due to YSB infestation and bolstering food security.

2 Life cycle of yellow stem borer

Yellow stem borer encompasses its life cycle through four stages viz, egg stage, larval stage, pupal stage, and adult stage [9]. Life cycle of YSB is presented in Fig. 1.

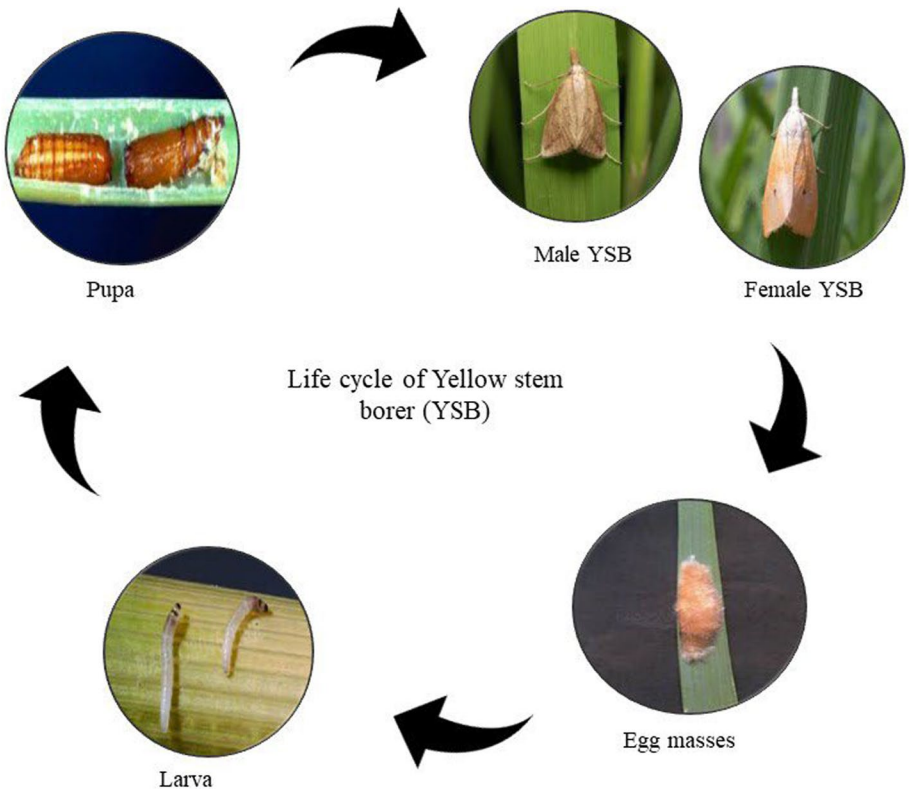


Fig. 1. Life cycle of Yellow Stem Borer (YSB) [10–12]

2.1 Egg stage

The eggs of YSB are creamy white, flattened, and oval (disc-like) in shape. They are laid in mass and covered with brownish hairs from the anal tufts of female moths [13]. They are laid mostly on the dorsal surface of the leaves [14]. A single female can lay 100–200 eggs and each egg mass consists of 50–80 eggs. The egg period is 5–8 days [15]. The life cycle of YSB is greatly influenced by temperature and relative humidity and is predicted to be decreased roughly by 2 days for every 1° rise in temperature. For a maximum number of eggs deposition, the optimum temperature is 29 °C with 90% RH. The optimum temperature for eggs to hatch is 24–29 °C and the relative humidity is 90–100% [16].

2.2 Larva

Larvae have a cream-colored body and reddish-brown head. The body length of the fully grown larvae is up to 20 mm long. They are yellow and white with a finely divided prothoracic shield [17]. The larval period is 28–30 days. The larva bores an exit hole and undergoes pupation usually at the base of the plant [9]. The larvae undergo four to six larval instar stages to become adults. The first instars are non-motile, and they do not migrate [17]. The second and fourth instar larvae show moderate migration, the third instar larvae migrate actively but the fifth instar larvae hardly move [18]. Second instar larvae enclose themselves in leaves making tubes and later detach themselves from the leaf. They fall on the water surface and get attached to the tiller and bore into the stem. The second, third, and fourth instar larvae are creamy white, fifth instar larva are dirty white and sixth instar larvae are greenish yellow to yellowish white [19].

2.3 Pupa

The pupal stage continues for 6 to 12 days. The pupa is light green in color. It forms a silky white cocoon and rests underneath the base of the plant and harvested plants [9].

2.4 Adult

The male and female of yellow stem borer are sexually dimorphic. The female has a wide abdomen, and their tip is covered with yellowish hairs forming a circular pattern, while, in males the abdomen is slender, and the anal end has a thin, hairy covering on the dorsal side. In females, the forewings are dark yellow to pale yellow from the tip or yellow with an orange-colored tinge and have a prominent black disc-like spot [9]. They have white hind wings with yellow shades in the coastal half. In males, the forewings are moderately orange-yellow with a dark color underside. They have pale and straw-colored hind wings with transparent and traceable major veins. In male, the wingspan is 20–30 mm and in the female wingspan is 24–36 mm [17].

The egg of YSB takes 5 to 8 days to hatch and becomes a larva. The larva after 28–30 days turns into a pupa and again after 8 to 10 days, the pupa turns into an adult [20]. The longevity of adult is 2–3 days [19]. Total life cycle of YSB happens to be 43 to 51 days.

3 Distribution of YSB

YSB is monophagous feeding only on paddy plants [21]. YSB is majorly found in overall tropical Asia and in temperate regions in areas where temperature remains above 10 °C and annual rainfall is more than 1000 mm. YSB is primarily found in Asian countries like Nepal, India, Pakistan, Sri Lanka, Afghanistan, China, Japan, Bangladesh, Thailand, Myanmar, Vietnam and many other countries[17]. They are mostly found in lowlands and flood-prone rice ecosystems [21]. Generally, they are found in leaf-over stubbles in aquatic environments.

4 Host plants of YSB

Yellow stem borers are monophagous pests of rice [22]. They feed only on rice plants.



Fig. 2. Death heart and white ear-head caused by YSB in rice

5 Nature of damage

After hatching from the eggs larvae of YSB enter the leaf sheath, it starts feeding uncontrollably for 2–3 days and then bores into the stem and starts staying in the pith and meanwhile, feeds on the inner surface of the walls. Generally, only one larva enters the plant [9] and 2–4 larvae per plant can be noticed occasionally [18]. Dispersal of larva generally occurs by ballooning (a process highly influenced by wind). Larvae themselves hang down by silken thread during dispersal and are blown to nearby plants, or they may fall into the water and swim freely until they reach rice plants [18]. When this activity of larvae occurs in the vegetative stage of growth in the plant, the central leaf whorl does not unfold, but starts turning brownish and finally dries off, but the lower leaves remain healthy and green and the symptom is known as dead heart (DH). In this condition, the affected tillers dry out and do not bear panicles [17]. The damage caused by stem borers is mostly visible at the time of rice harvest. They cause the rice panicles to turn white due to empty grains and the condition is called whitehead [23]. The yellow stem borer causes dead hearts in the vegetative stage and white ear-heads at the reproductive stage [24]. Tiller injury during the reproductive phase not only reduces grain yield by resulting in whiteheads but also lowers the grain weight of healthy panicles [25]. The highest infestation of YSB larvae occurs in flowering stages and lowest in the panicle initiation stage of growth. [15] reported the derived yield loss of 1% from dead heart, 19% from white earhead, and over 23% due to both dead heart and white earhead. White ear-head (WE) and dead heart incidence in YSB were negatively related to mean temperature and positively related to relative humidity and rainfall [26].

The YSB attacks the rice plants from the seedling to the harvesting stage till the affected tillers are completely lost [27]. The incidence of stem borers to the extent of 20% was recorded in early sown crops and 80% in late sown rice crops [28]. Therefore, late-planted crops are more affected by YSB than early-planted crops [27]. Dead heart and white ear-head caused by YSB are shown in Fig. 2.

6 Economic threshold level (ETL) and economic injury level (EIL)

6.1 Economic threshold level (ETL)

The economic threshold level can be defined as the density of pests at which control measures should be implemented to prevent the outbreak of the pest population from reaching the economic injury level. the concept of ETL is often applied to control insects and is a preventive management strategy [29]. No action is required as long as the pest density stays below this threshold; however, if the insect population density ascends over this point, a control measure is recommended. Therefore, the economic threshold is intended to prevent a pest population from spreading to the point where

its damages result in monetary losses that are equivalent to the cost of control, which is the Economic Injury level (EIL) [30]. The ETL for the control of YSB is 5%-10% dead-hearts with 2% white ear [31].

6.2 Economic injury level

Economic injury level (EIL) is a key concept in crop production because it quantifies the cost-benefit ratio and guides all the pest control decisions in sustainable agriculture and integrated pest management (IPM). EIL is the lowest pest population that causes economic damage [32]. In the agricultural production system, EIL is considered as the point where the benefits of pest management activities are equal to the costs of pest management [33]. The EIL of stem borers can occur above 10% dead-hearts [34].

7 Integrated pest management (IPM) of YSB

Integrated pest management (IPM) is an approach to control insect pests and prevent their damage to the crop. The IPM practice generally includes the biological, physical, and cultural methods for effective and environmentally safe pest control [35]. IPM provides cost-effective, environmentally friendly, and socially acceptable methods of managing insects and pests [6]. IPM is a key approach, and if implemented properly, it will lead to much improved control of the stem borers. YSB can be managed in the incidence phase using IPM [36]. Various learning activities like Farmers Field School (FFS) and Focus Group Discussions (FGD) assist farmers in making timely and suitable decisions on crop management, which are beneficial in the implementation of integrated pest management [37].

7.1 Cultural practices for the management of YSB

7.1.1 Use of resistant varieties

Resistant and tolerant varieties play a key role in the reduction of yield losses. Host-plant resistance is a key component in the integrated control of rice insect pests [38]. Stem borer resistance may result from tolerance (the capacity to adapt and recover from damage), antibiosis (increased mortality, reduced longevity and reproduction of insect), and antixenosis (reduced plant attractiveness). The majority of rice varieties result in slower larval development, smaller body size and higher larval mortality [23]. Many rice varieties are resistant to YSB. Rice varieties like Sarju 52 and Silky Mansuli are resistant to yellow stem borer attack [39]. For effective pest control, varietal selection is very necessary [40]. If the resistant varieties are absent, early-maturing varieties should be selected for cultivation since, they may escape the damage of late-coming pests [17].Varieties like Hardinath-1, Ram dhan, Mithila and Loktantra are early maturing varieties [41]. ADT 44, ADT 47, Ajay, Anjali, JKRH 401, Mahanda, Surya are some Indian rice varieties resistant to YSB [17]. The use of these varieties can lower the YSB infestation.

7.1.2 Proper use of nitrogenous fertilizer

Excessive use of nitrogenous fertilizer should be avoided since a high level of nitrogen attracts stem borers. For the reduction of damage by pests, nitrogenous fertilizers should be applied in three split doses [42].

7.1.3 Early sowing

Early sowing of rice crops reduces the YSB infestation since at that time the natural enemies and *Trichogramma* spp. are abundantly present as compared to normal sown and late sown rice. In the case of late-sown crops the entire pest would be feeding the late sown rice since the surrounding rice crops might have completed their susceptible stages [43].

7.1.4 Cultivation of direct seeded rice (DSR)

YSB is found in fields with a lot of water. The second instar larvae encase themselves in body leaf wrappings to form tubes and descend to the water surface after removing the leaf. Afterward, they cling to the tillers and bore into the stem. Hence, the greater number of YSB is seen in deep-water rice than in direct-seeded rice. So, the cultivation of DSR helps to reduce YSB infestation [44].

7.1.5 Clipping of seedlings

The female YSB lay eggs in the upper region of the plant on the abaxial surface, thus clipping of tip of seedlings are crucial to remove YSB eggs in early stage of rice planting or before transplanting into main field [45].

7.1.6 Transgenic rice

Transgenic rice plants incorporated with *Bt* genes (*Bacillus thuringiensis* Berliner) are resistant to YSB and they can be prepared by the introduction of crystal insecticidal protein (δ-endotoxin) genes or cry gene proteins from *Bt* into rice plants by using genetic engineering. The cry genes in the transgenic rice plants are highly toxic to YSB larvae and help to reduce the damage caused by their feeding [21]. The use of *Bt* genes in plants showed 100% eradication of the stem borer since they produce toxins to kill the pests [46]. YSB can develop resistance to such proteins over time, therefore intercropping non *Bt* rice with *Bt* rice helps to slow down resistance while maintaining susceptibility. Combining *Bt* rice with additional pest management strategies (for example, natural enemies like *Trichogramma*) might improve efficacy [47]. However, the development of *Bt* incorporated transgenic rice is not popular, and it has not yet received any direct advantages of development because some Nepalese people feel *Bt* rice might be harmful to human health and the environment. Also, the scientific community recommends approving the introduction of *Bt* rice only if the danger is minimal or extremely low [48]. Further research on the development of transgenic rice, their efficacy, and sustainability should be done in the coming days.

7.2 Mechanical control methods for the management of YSB

The common methods like hand picking and destroying the larvae, eggs and infested stems are effective for small-scale farming. The adult stem borers can be monitored and captured by using traps like pheromone traps, Light traps and sticky traps [20]. Pheromone traps monitor the male yellow stem borer, trap them and disrupt their mating with female moths. Light traps are used to monitor and control YSB with white and ultraviolet color light sources [49]. Female moths are attracted to light and the use of light traps helps to reduce their population number [50].To disrupt mating, sex pheromone of YSB consisting of (Z)-9-hexadecenal and (Z)-11-hexadecenal in 1:3 ratio can be used [17]. Also, powder, wax-based Specific pheromone and lure application technology (SPLAT), and talc as mating disruption tools against the yellow stem borer, are economical to constantly disrupt the mating process completely [51].

7.3 Biological control methods of YSB

Biological control is considered to be one of the most effective and environmentally friendly management practices [52]. It involves the use of natural enemies like predators, parasites and parasitoids of the pest [42]. Spiders and beetles are the common predators of YSB. Beetle *Paederus* sp. (Coleoptera: Staphylinidae) and spiders: *Atypena* sp. (Araneae: Linyphiidae), *Araneus* sp. (Araneae: Araneidae), *Lycosa* sp. (Araneae: Lycosidae), *Plexippus* sp. (Araneae: Salticiidae) and *Oxyopes* sp. (Araneae: Oxyopidae) are the main predator species found [53]. In natural conditions, YSB is attacked by large number of predators which eases its control so it is really necessary to conserve its natural enemies around the rice fields [17]. Some of the important parasitoids of YSB are listed in Table 1.

Parasitoids play a vital role in controlling the population and damage intensity of yellow stem borer [53]. Field releases of *Trichogramma japonicum* and *T. chilonis* in the field can control the YSB biologically [54]. In Cuijia, China, *T. chilonis* was

Table 1. Important parasitoids of YSB: (Source: [17])

Scientific name	Common name	Order	Family	Target stage
Cotesia flavipes	Braconid wasp	Hymenoptera	Braconidae	Larva
Elasmus albopictus	Plasmid wasp	Hymenoptera	Eulophidae	Larva
Eriborus sinicus	Ichneumonid wasp	Hymenoptera	Ichneumonidae	Egg
Telenomus dignus	Scelionid wasp	Hymenoptera	Scelionidae	Egg
Tetrastichus schoenobii	Eulophid wasp	Hymenoptera	Eulophidae	Pupa
Trichogramma japonicum	Trichogrammatid	Hymenoptera	Trichogrammatidae	Egg
Trichogrammatoidea nana	Trichogrammatid	Hymenoptera	Trichogrammatidae	Egg

found to have higher parasitism of YSB than any other strains [55]. To increase the effectiveness of parasitoids and predators in rice field, the natural enemies present in the field should be conserved by creating a favorable environment for them. Pesticide use should be avoided since it might kill natural enemies [17]. Refugia plant like marigolds, sunflowers, coriander, etc provides a source of feed for natural enemies, especially to parasitoids so, the plantation of these plants near rice fields helps to conserve them [56]. Refugia plants can be implemented in field borders, intercropping, rotational cropping, or mixed planting with rice.

Biopesticides like Neemazal @ 500 ml/ha, Lemon grass oil @ 500 ml/ha, *B. thuringiensis* @ 1–2 l/ha, Azadirachtin @ 0.5–3 l/ha depending on pest pressure, and entomopathogenic fungus like *Metarhizium anisopliae* @ 2 kg/ha wettable powder, *Beauveria bassiana* @ 2 kg/ha wettable powder can be incorporated into the IPM program for the control of YSB [57–59].

In Nepal, biological control is gaining popularity, but many farmers are still unfamiliar with these methods and rely on chemical pesticides [60]. To promote biopesticide and beneficial insects such as parasitoids and predators, farmers need to be better equipped to utilize these tools and be further educated about their benefits. Supportive policies, enhanced research in integrated pest management methods, and conservation efforts are essential to protect these natural enemies and promote sustainable agricultural practices [61, 62].

7.4 Chemical control of YSB

Several insecticides have been identified for the control of rice stem borer, but the improper use of chemical pesticides can be hazardous and may accumulate into harvested products[63]. To effectively manage YSB, chemical pesticides are applied to rice at the seedling and reproductive phases[64]. Judicious use of pesticides can be considered if the pest population reaches the economic threshold level. However, it is important to use selective insecticides that are friendly to their natural enemies, and recommended application guidelines should be followed. Spray formulation should be used against YSB when the crop reaches the maximum tillering stage since granules are not highly effective. Spraying of Flubendiamide 20% WG 50 g/ac, Fipronil 5% SC 400–600 g/ac, Chlorantraniliprole 18.5% SC 60 ml/ac, Chlorantraniliprole 0.4% G 4 kg/ac, or Thiamethoxam 25% WG 40 g/ac is recommended for the effective chemical control of yellow stem borer [20]. Chlorantraniliprole (18.5 SC), Chlorpyriphos (20 EC), Lambda cyhalothrin (5 EC), Fipronil (5 EC), Emamectin benzoate (5 SG), and Cartap hydrochloride (50 SP), are the commonly used pesticides by farmers in Nepal to control YSB but these are harmful to predators and beneficial insects also not compatible with biological controls. Among these insecticides, chlorantraniliprole 18.5 SC was found to be the most effective in minimizing dead heart (DH%) symptoms, reducing them from 5.93% to 4.46%, and minimizing white heads from 7.56% to 2.5% in comparison to the control treatment [65].

8 Conclusion

Yellow stem borer *Scirpophaga incertulas*, is regarded as one of the destructive pests to rice crops worldwide. The control of yellow stem borer (YSB) in rice cultivation in Nepal is critical to ensure the production of rice and food security. Integrated pest management (IPM) solutions, which include cultural, biological, mechanical, and chemical approaches, are critical for long-term YSB control. Farmers should focus on these methods for the control of pests since these methods help to preserve the soil and environmental health, playing a significant role in sustainable agriculture. Pesticides to control YSB should be only applied when the population number reaches the economic threshold level (ETL) and when no biological control is conducted [20]. While the use of selective pesticides such as chlorantraniliprole and fipronil is still an option, teaching farmers optimal application time reduces environmental effects and preserves beneficial creatures. This integrated strategy enables long-term YSB management while preserving environmental balance in rice growing systems.

Implementation of suitable strategies to control YSB in time leads to better production of rice and contributes to global food security. All these components like the use of biopesticides, mating disruption, and agroecological measures like refugia plants are all important components of sustainable pest management strategies. They provide environmental advantages, increase biodiversity, have minimal impact on non-target organisms, and lessen the need for chemical pesticides. Enhancing Farmer Field Schools, investing in research for biological control agents and resistant rice varieties, strengthening agricultural extension services, and enacting supporting policies to encourage IPM adoption are all important. Encouraging collaboration among farmers, academics, and policymakers will increase the adoption of these sustainable practices, resulting in better pest control and environmental protection in Nepal and other YSB affected regions.

Author contribution SS conceived the idea for the review article, drafted the initial version of the article. AKB conducted an extensive literature review, analyzed compiled information, reviewed and edited the article. ST collaborated on gathering the relevant information, critically reviewed the article. KT provided supervision and guidance throughout the process, provide final approval for submission.

Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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